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NULL POLYNOMIALS OVER A FINITE RING NEED NOT FORM A TWO-SIDED IDEAL

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Abstract. We give an example of a finite ring for which the set of polynomials inducing the zero function fails to be a two-sided ideal, disproving a conjecture of Werner.

1. Introduction

Let R be a unital ring, and let $R[x]$ denote the polynomial ring over R in a central indeterminate x. Let $\mathrm{N}(R) = \{f \in R[x] \mid f(r) = 0 \text{ for all } r \in R\}$ be the set of null polynomials with coefficients in R. Here we use the convention of right substitution, i.e., for $f = \sum_{i=0}^{n} a_i x^i$ and $r \in R$, we define $f(r) = \sum_{i=0}^{n} a_i r^i$.

If R is a commutative ring, then $\mathrm{N}(R)$ is easily seen to be an ideal of $R[x]$. However, if R is noncommutative, whether this holds is not clear, since evaluation at r fails to preserve multiplication in general. Nevertheless, the set $\mathrm{N}(R)$ is always a left ideal. Indeed, it is clearly an additive subgroup of $R[x]$. Moreover, if $g = \sum_j b_j x^j$ is an arbitrary polynomial and $f \in \mathrm{N}(R)$, then for all $r \in R$ we have $(gf)(r) = \sum_j b_j f(r) r^j = 0$. Werner proved that $\mathrm{N}(R)$ is a two-sided ideal for a large class of rings, including local rings, semisimple rings, matrix rings over arbitrary commutative rings, and rings of odd order [4, Theorem 3.7], and conjectured that this holds for all finite rings.

Further evidence for this conjecture was given by Frisch, who showed that $\mathrm{N}(R)$ is a two-sided ideal when R is the ring of upper triangular matrices over a commutative ring [2, Theorem 5.2]. More generally, the same conclusion holds when R is a structural matrix ring [3].

We disprove the conjecture by giving an example of a finite ring whose null polynomials do not form a two-sided ideal.

2. The counterexample

Throughout, $\mathrm{M}_n(\mathbb{F}_2)$ denotes the ring of $n \times n$ matrices over $\mathbb{F}_2$, and $\mathrm{T}_n(\mathbb{F}_2)$ its subring of upper triangular matrices.

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Theorem 2.1. *Let*

$$R = \left\{ \begin{pmatrix} A & B \\ 0 & A \end{pmatrix} \;\middle|\; A \in \mathrm{T}_2(\mathbb{F}_2) \text{ and } B \in \mathrm{M}_2(\mathbb{F}_2) \right\}.$$

Then R is a finite ring and $\mathrm{N}(R)$ is not a right ideal of $R[x]$. Consequently, $\mathrm{N}(R)$ is not a two-sided ideal of $R[x]$.

Proof. The set R is an additive subgroup of $\mathrm{M}_4(\mathbb{F}_2)$ and contains the identity matrix. Moreover,

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix} \begin{pmatrix} C & D \\ 0 & C \end{pmatrix} = \begin{pmatrix} AC & AD + BC \\ 0 & AC \end{pmatrix} \in R.$$

Thus R is a unital subring of $\mathrm{M}_4(\mathbb{F}_2)$. Equivalently, R is the trivial extension of $\mathrm{T}_2(\mathbb{F}_2)$ by the bimodule $\mathrm{M}_2(\mathbb{F}_2)$.

Let $E = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) \in \mathrm{T}_2(\mathbb{F}_2)$, and $G = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \in \mathrm{T}_2(\mathbb{F}_2)$. Further, let $P = \left(\begin{smallmatrix} E & 0 \\ 0 & E \end{smallmatrix}\right) \in R$, and $Q = \left(\begin{smallmatrix} G & 0 \\ 0 & G \end{smallmatrix}\right) \in R$. We claim that the polynomial

$$f = Qx^2 + Px^3 + (P + Q)x^4 + Px^5 + Px^6$$

satisfies $f \in \mathrm{N}(R)$ but $fP \notin \mathrm{N}(R)$.

We first show that $f \in \mathrm{N}(R)$. Let $r \in R$ be arbitrary and observe that

$$f = (Q + Px + Px^2)(x^2 + x^4).$$

Let $v = Q + Px + Px^2$ and $w = x^2 + x^4$. Although evaluation is not multiplicative in general, in this instance we do have $f(r) = v(r)w(r)$ since the coefficients of w are central. Now let

$$r = \begin{pmatrix} A & B \\ 0 & A \end{pmatrix} \in R, \qquad A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}.$$

Then

$$r^2 = \begin{pmatrix} A^2 & AB + BA \\ 0 & A^2 \end{pmatrix}.$$

Since $a, b, c \in \mathbb{F}_2$, we have

$$A^2 = \begin{pmatrix} a & b(a + c) \\ 0 & c \end{pmatrix}.$$

Thus

$$A^2 = A \quad \Longleftrightarrow \quad b = 0 \text{ or } a \neq c.$$

Consequently, either $A^2 = A$, or $b = 1$ and $a = c$. In the latter case,

$$A = \varepsilon I_2 + G \qquad \text{for some } \varepsilon \in \mathbb{F}_2,$$

where I_2 denotes the 2×2 identity matrix.

Suppose first that $A^2 = A$. In this case,

$$r^2 = \begin{pmatrix} A & AB + BA \\ 0 & A \end{pmatrix},$$

and

$$\begin{aligned} r^4 &= \begin{pmatrix} A & AB + ABA + ABA + BA \\ 0 & A \end{pmatrix} \\ &= \begin{pmatrix} A & AB + BA \\ 0 & A \end{pmatrix} \\ &= r^2, \end{aligned}$$

where we have used that the characteristic is 2. Thus $w(r) = r^2 + r^4 = 0$, which implies $f(r) = 0$.

Now suppose that $A = \varepsilon I_2 + G$. Since $G^2 = 0$, we have $A^2 = \varepsilon I_2$ and hence

$$r^2 = \begin{pmatrix} \varepsilon I_2 & AB + BA \\ 0 & \varepsilon I_2 \end{pmatrix}, \quad \text{and} \quad r^4 = \begin{pmatrix} \varepsilon I_2 & 0 \\ 0 & \varepsilon I_2 \end{pmatrix}.$$

Consequently,

$$w(r) = r^2 + r^4 = \begin{pmatrix} 0 & AB + BA \\ 0 & 0 \end{pmatrix}.$$

Recall that because w has central coefficients,

$$f(r) = Qw(r) + Prw(r) + Pr^2w(r).$$

Setting $C := AB + BA$, a block-matrix multiplication gives

$$f(r) = \begin{pmatrix} 0 & GC + EAC + \varepsilon EC \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & GC + EGC \\ 0 & 0 \end{pmatrix},$$

because $A = \varepsilon I_2 + G$. Finally, since $EG = G$, the upper-right block is

$$GC + EGC = GC + GC = 0.$$

Thus $f(r) = 0$ also in this case. Since $r \in R$ was arbitrary, we conclude that $f \in \mathrm{N}(R)$.

It remains to show that $fP \notin \mathrm{N}(R)$. Using $P^2 = P$ and $QP = 0$, we obtain $fP = Px^3 + Px^4 + Px^5 + Px^6$. Now let $J = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right) \in R$. Then $J^3 = \left(\begin{smallmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$, while all higher powers vanish. Hence

$$(fP)(J) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq 0.$$

This shows that $\mathrm{N}(R)$ is not closed under right multiplication by P, so in particular $\mathrm{N}(R)$ is not a right ideal of $R[x]$. □

Remark 2.2. Although R has a prescribed zero lower-left block, it is not a structural matrix ring: membership in R also requires its two diagonal blocks to be equal, a condition that cannot be expressed solely by prescribing certain entries to be zero. Hence the known results for structural matrix rings do not apply here.

Remark 2.3. This example can also be used to construct a finitely generated torsion-free $\mathbb{Z}$-algebra for which the set of integer-valued polynomials does not form a ring. This provides the exampled requested in [1, Problem 27a], and disproves the conjecture in [1, Problem 27b].

Indeed, replace $\mathbb{F}_2$ with $\mathbb{Z}$ in the definition of the ring R to obtain the module-finite, free $\mathbb{Z}$-algebra S. Then $R \cong S/2S$. By [4, Theorem 2.4], the failure of $\mathrm{N}(R)$ to be a two-sided ideal implies that the integer-valued polynomials over S do not form a ring.

Declaration on the use of LLMs

The counterexample and an initial proof were generated by GPT-5.6 Sol. The author subsequently verified the construction independently and simplified the proof. The author takes full responsibility for the mathematical content of the article.

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